B. Scenes From a Memory

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1000 | Tags: brute force, constructive algorithms, implementation, math, number theory |
https://codeforces.com/problemset/problem/1562/B

During the hypnosis session, Nicholas suddenly remembered a positive integer n, which **doesn't contain zeros in decimal notation**.

Soon, when he returned home, he got curious: what is the maximum number of digits that can be removed from the number so that the number becomes **not prime**, that is, either composite or equal to one?

For some numbers doing so is impossible: for example, for number 53 it's impossible to delete some of its digits to obtain a not prime integer. However, **for all n in the test cases of this problem, it's guaranteed that it's possible to delete some of their digits to obtain a not prime number**.

Note that you cannot remove all the digits from the number.

A prime number is a number that has no divisors except one and itself. A composite is a number that has more than two divisors. 1 is neither a prime nor a composite number.

Input

Each test contains multiple test cases.

The first line contains one positive integer t ($1 \le t \le 10^3$), denoting the number of test cases. Description of the test cases follows.

The first line of each test case contains one positive integer k ($1 \le k \le 50$) — the number of digits in the number.

The second line of each test case contains a positive integer n, which doesn't contain zeros in decimal notation ($10^{k-1} \le n$ lt 10^k). It is guaranteed that it is always possible to remove less than k digits to make the number not prime.

It is guaranteed that the sum of k over all test cases does not exceed 10^4.

Output

For every test case, print two numbers in two lines. In the first line print the number of digits, that you **have left** in the number. In the second line print the digits left after all delitions.

If there are multiple solutions, print any.

Example

Input

```
7
3
237
5
44444
3
221
2
35
3
773
1
4
30
6262216262216262216262216262221
```

Output

```
2
27
1
4
1
1
2
35
2
77
1
4
1
6
```

Note

In the first test case, you can't delete 2 digits from the number 237, as all the numbers 2, 3, and 7 are prime. However, you can delete 1 digit, obtaining a number $27 = 3^3$.

In the second test case, you can delete all digits except one, as $4 = 2^2$ is a composite number.